Review plan · for the teacher

Data science in the classroom: unit check

Data science · Level 1 · Unit check

Time	Materials
40 min	Board, chalk, paper. No devices needed.

Learning goals

- The data science loop
- Middles and spread
- Charts that fit and tell the truth
- Careful claims and private pupils

Activity: Team quiz

1. Teams (2 min): Make teams of four and give each team a sheet for its answers.
2. Quiz (20 min): Read each question aloud; each team agrees on one answer and writes it down.
3. Explain (10 min): Reveal each answer and ask a team to explain it with an example from the unit.
4. Wrap-up (8 min): Work through the question most teams got wrong, together on the board.

Discussion

- Which idea from this unit will you use first?
- Which school record will you explore with your class?
- What will you change about sharing results?

Exit ticket

- One habit I will use this week, and where.

Try it in class tomorrow
Pick one school record, ask one measurable question of it with your class, and report the answer with a middle, a spread and one honest chart, without naming any child.

Quiz sheet · for each team

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Team _____ Date _____

1 **Which is the best data question for a class to investigate?**

◯ Do pupils like our school? ◯ How many pupils walk to school each day this week? ◯ Is walking good?

2 **Put the data science loop in order.**

☐ Explore
☐ Ask
☐ Explain
☐ Collect

3 **Complete the sentence.**

For 2, 3, 3, 4 and 8, the median is _____ and the mean is _____ .

Word bank: 8 · 3 · 4 · 6

4 **One extreme value moves the median more than the mean.**

◯ True ◯ False

5 **Match each question to its chart.**

Attendance week by week	• • • •	Scatter chart
Pupils per class	• • • •	Line chart
Height and arm span of each pupil	• • • •	Bar chart

6 **A bar chart's axis starts at 95. What does that do?**

◯ Nothing: axes don't matter ◯ Makes small differences look big ◯ Makes the chart more accurate

7 **The range of a set of marks is 0. Which must be true? Choose all that apply.**

☐ Every mark is the same ☐ The mean equals the median ☐ Every pupil scored 0

8 **Pupils who own more books get better marks, so giving every pupil books will raise marks.**

◯ True ◯ False

9 **Which is the safest way to share a test's results with parents?**

◯ A ranked list in the class WhatsApp group ◯ Each pupil's mark by name on the wall
◯ The class median and range, with each family told its own child's mark privately

10 **Match each term to its meaning.**

Mean	• • • •	Highest minus lowest
Median	• • • •	The total shared out equally
Range	• • • •	Two things that go together
Correlation	• • • •	The middle value in order

Answers · for the teacher

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1 **How many pupils walk to school each day this week?**

It is clear and can be counted each day.

2 **Top to bottom: 3, 1, 4, 2**

A question decides what to collect; exploring comes before explaining.

3 **For 2, 3, 3, 4 and 8, the median is 3 and the mean is 4.**

In order, the middle (third) value is 3.

4 **False**

The mean uses every value, so an extreme value pulls it; the median only depends on the middle position.

5 **Attendance week by week — Line chart**
Pupils per class — Bar chart
Height and arm span of each pupil — Scatter chart

Time needs a line, groups need bars, and two measurements of each pupil need a scatter chart.

6 **Makes small differences look big**

Bars from 95 show only the tops of the values, so a difference of 2 can look several times bigger.

7 **Every mark is the same; The mean equals the median**

Highest minus lowest is 0 only when all marks are equal, and then the mean and median are that same mark, which need not be 0.

8 **False**

The pattern may come from something else, such as families with more time to help with homework.

9 **The class median and range, with each family told its own child's mark privately**

The class picture informs everyone; each child's mark stays between the school and that family.

10 **Mean — The total shared out equally**
Median — The middle value in order
Range — Highest minus lowest
Correlation — Two things that go together

These four words carry most of this unit: two middles, one spread, and a pattern that is not yet a cause.